Question 1

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You design a standard 2-layer fully connected neural network. All activations are sigmoids and your optimizer is stochastic gradient descent. You initialize all the weights and biases to zero and forward propagate an input $\mathbf{x}$ in the network. What is the output $\hat{y}$ of the network?

Select one:

- a. 0
- b. $e^{\mathbf{x}}$
- c. 0.5
- d. -1
- e. 1

- **2-layer** fully connected neural network
- **Sigmoid** activations
- **Weights and biases initialized to 0**
- **Optimizer and training don't matter yet — we only do a forward pass**
- Input: some vector x

🔁 Forward pass:

1 First layer pre-activation:

$$z^{(1)} = W^{(1)}x + b^{(1)} = 0 \cdot x + 0 = 0$$

2 First layer activation (sigmoid):

$$a^{(1)} = \sigma(z^{(1)}) = \sigma(0) = \frac{1}{1+e^{-0}} = 0.5$$

3 Second layer pre-activation:

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)} = 0 \cdot 0.5 + 0 = 0$$

4 Second layer activation (sigmoid again):

$$a^{(2)} = \sigma(z^{(2)}) = \sigma(0) = 0.5$$

If all weights are initialized to zero, **every neuron computes the same output**, and all gradients become the same — leading to **no learning** (symmetry breaking fails). That's why modern networks use random initialization (e.g. Xavier, He).

Question **2**

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Consider the model defined in the question before with parameters initialized with zeros. $\mathbf{W}^{[2]}$ denotes the weight matrix of the second layer. You forward propagate a batch of examples, and then backpropagate the gradients and update the parameters. Which of the following statements is true?

Select one:

- a. The entries of the weight matrix $\mathbf{W}^{[2]}$ are all positive.
- b. The entries of the weight matrix $\mathbf{W}^{[2]}$ may be positive or negative.
- c. The entries of the weight matrix $\mathbf{W}^{[2]}$ are all negative.
- d. The entries of the weight matrix $\mathbf{W}^{[2]}$ are all zeros.

Clear my choice

I think b) should be correct. I think not all zeros, unless mean target = 0.5. But not sure.

Question **5**

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You are given an input $\mathbf{x} = \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}$ and a kernel weight matrix $\mathbf{w} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. What is the result of the **transpose convolution operation** $*^T$ (stride 2) of

$$\mathbf{w} *^T \mathbf{x} = ?$$

Select one:

- a. $= \begin{pmatrix} 0 & 1 & -1 \\ 2 & -1 & -1 \\ -4 & -2 & 6 \end{pmatrix}$
- b. $= \begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix}$
- c. $= \begin{pmatrix} 0 & 1 & 0 & -2 \\ 2 & 3 & -4 & -6 \\ 0 & -1 & 0 & 2 \\ 2 & 3 & -4 & -6 \end{pmatrix}$
- d. $= \begin{pmatrix} 0 & 2 & 1 \\ 6 & -14 & 6 \\ 2 & 3 & 0 \end{pmatrix}$
- e. $= \begin{pmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & -2 \\ 2 & 2 & 4 & 4 \\ 2 & 2 & -4 & -4 \end{pmatrix}$
- f. $= (3)$
- g. $= (42)$
- h. $= \begin{pmatrix} 8 & -5 & 2 \\ -11 & 5 & 13 \\ 19 & 21 & -29 \end{pmatrix}$

Clear my choice

Question 15

Answer saved

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Suppose that you have a fixed, multi-dimentional, non-Gaussian distribution $p(\boldsymbol{x})$ that you want to approximate with a Gaussian distribution. You decide that you want to approximate this distribution with a Gaussian $q(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with some mean vector $\boldsymbol{\mu}$ and some covariance matrix $\boldsymbol{\Sigma}$. You now decide to minimize the Kullback-Leibler divergence between the two distributions $\mathrm{KL}(p \,||\, q)$ with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.

Which expressions for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ mimimize the KL-divergence between the two distributions?

(The usual notation of the lecture is used: $\mathbb{E}$ is expectation, $\mathbb{C}\mathrm{ov}$ is the covariance, $\boldsymbol{I}$ is the identity matrix)

- [ ] a. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x}^T\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \boldsymbol{I}$
- [x] b. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \mathbb{C}\mathrm{ov}(\boldsymbol{x})$
- [ ] c. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x}^T\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\boldsymbol{x}^T)$
- [ ] d. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \boldsymbol{I}$
- [x] e. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\boldsymbol{x}\boldsymbol{x}^T)$

After taking a look at gpt solution and previous discussions in the chat I think e) should be wrong and only b) correct.

❌ Why the other marked answer (e) is **wrong**:

❌ e. $\mu = \mathbb{E}[x]$, $\Sigma = \mathbb{E}[xx^T]$

- $\mathbb{E}[xx^T]$ is the **second moment**, **not** the **covariance**.
- Covariance is:

$$\mathrm{Cov}(x) = \mathbb{E}[xx^T] - \mathbb{E}[x]\mathbb{E}[x]^T$$

- So **option e** overestimates the covariance and gives the wrong spread.

Question 21

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The main difference between the frequentist and Bayesian approach is marginalization over parameters when we define the probability for a class label given the input $\mathbf{x}$ and dataset $\mathcal{D}$. To give a more formal definition:

$$p(\mathbf{y}^{test}|\mathbf{x}^{test}, \mathcal{D}) = \mathbb{E}_{\mathbf{w}\sim p(\mathbf{w}|\mathcal{D})}\left[p(\mathbf{y}^{test}|\mathbf{w}, \mathbf{x}^{test})\right]$$

Deep Ensembles try to approximate the full expectation with (weighted) averages. Which of this options are valid?

$\mathbb{E}_{\mathbf{w}\sim p(\mathbf{w}|\mathcal{D})}\left[p(\mathbf{y}^{test}|\mathbf{w}, \mathbf{x}^{test})\right] \approx$

- [x] a. $\approx \frac{1}{M}\sum_{m=1}^{M} p(\mathbf{y}^{test}|\mathbf{w}_m, \mathbf{x}^{test})$, where $\mathbf{w}_m$ denotes the weights of the m-th network of the ensemble.
- [ ] b. $\approx p(\mathbf{y}^{test}|\mathbf{w}, \mathbf{x}^{test})$
- [x] c. $\approx \frac{1}{M}\sum_{m=1}^{M} \alpha_m p(\mathbf{y}^{test}|\mathbf{w}_m, \mathbf{x}^{test})$ with $\alpha_m = \mathrm{softmax}(\mathbf{z})_m$ for some random normal distributed vector $\mathbf{z}$
- [x] d. $\approx \frac{1}{M}\sum_{m=1}^{M} p(\mathbf{y}_m^{test}|\mathbf{w}, \mathbf{x}_m^{test})$ where the subscript m of the test data denotes the m-th equally sized fraction of the full data.
- [ ] e. $\approx \sum_{m=1}^{M} p(\mathbf{y}_m^{test}|\mathbf{w}, \mathbf{x}_m^{test})$ where the subscript m of the test data denotes the m-th equally sized fraction of the full data.
- [ ] f. $\approx \sum_{m=1}^{M} p(\mathbf{y}^{test}|\mathbf{w}_m, \mathbf{x}^{test})$

c) also probably also correct

15.1. Deep Ensembles

- Main idea: approximate expectation with averages

$$p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{x}^{\text{test}}, \mathcal{D}) = \mathbb{E}_{\boldsymbol{w} \sim p(\boldsymbol{w}|\mathcal{D})} \left[p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{w}, \boldsymbol{x}^{\text{test}}) \right]$$

- Train several DNNs with different random seeds, obtain parameters $\boldsymbol{w}^{(m)}$ and take average

$$p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{x}^{\text{test}}, \mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^{M} p(\boldsymbol{y}^{\text{test}} \mid \boldsymbol{w}^{(m)}, \boldsymbol{x}^{\text{test}},)$$

- Averages can be weighted differently
- Appears as simple technique, but performs well in practice

Question 22

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Assume that you want to implement a new layer for a deep neural network that employs the following transformation

$$\boldsymbol{h}^{(l+1)} = \underbrace{\boldsymbol{M}\text{softmax}(\beta\, \boldsymbol{M}^T\, \boldsymbol{h}^{(l)})}_{:=\boldsymbol{p}},$$

where $\boldsymbol{h}^{(l)} \in \mathbb{R}^d$ is the input to this layer l, β is a positive hyperparameter to be set by the user, $\boldsymbol{M} \in \mathbb{R}^{d \times J}$ is a matrix of J patterns with d features provided by the domain expert, and $\boldsymbol{p}$ is defined as $\boldsymbol{p} := \text{softmax}(\beta\, \boldsymbol{M}^T\, \boldsymbol{h}^{(l)})$. Which of the following statements about this layer are true?

- [x] a. $\boldsymbol{h}^{(l+1)}$ is equivariant to permutations of the columns of $\boldsymbol{M}$.
- [x] b. $\boldsymbol{p}$ is invariant to permutations of the columns of $\boldsymbol{M}$.
- [ ] c. $\boldsymbol{p}$ is invariant to permutations of the columns of $\boldsymbol{M}$.
- [ ] d. $\boldsymbol{p}$ is equivariant to permutations of the columns of $\boldsymbol{M}$.
- [x] e. $\boldsymbol{p}$ is equivariant to permutations of the rows of $\boldsymbol{M}$.
- [ ] f. $\boldsymbol{h}^{(l+1)}$ is invariant to permutations of the columns of $\boldsymbol{M}$.
- [ ] g. This layer cannot be built in deep learning architectures because gradients, i.e. $\frac{\partial \boldsymbol{h}^{(l+1)}}{\partial \boldsymbol{h}^{(l)}}$, cannot be backpropagated

22 d and f correct.

d) Matrix multiplication is equivariant to the permutation of rows of the first matrix. But M is transposed. Softmax doesn't change the order.

f) It is invariant because you change the order of columns of M and the order of rows of p changes accordingly so in the end you have the same result

Question 23

Answer saved

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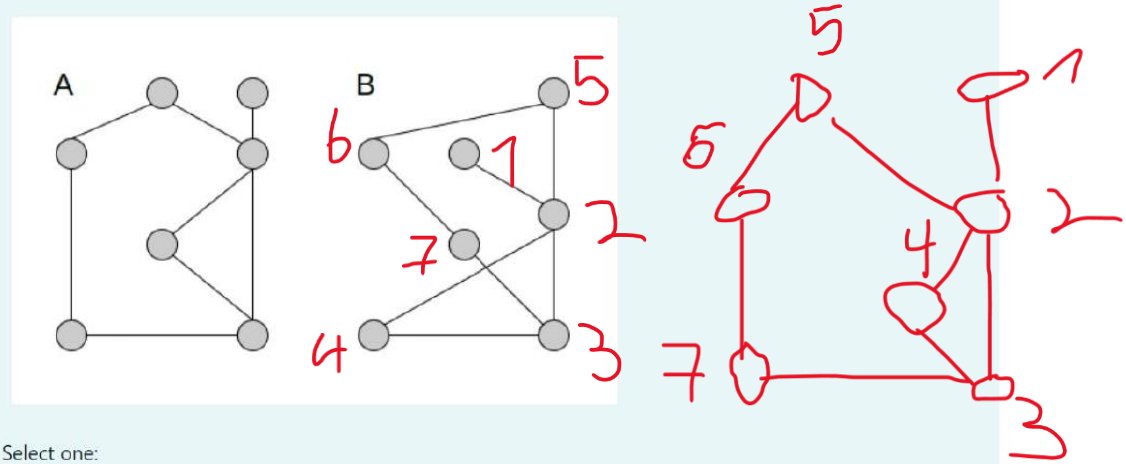
Flag question

Do the following two graphs fulfill the necessary condition for graph isomorphism, the Weisfeiler-Lehman test?



Select one:

- True
- False